4:15

12.1 Elliptic Partial Differential Equations

(p.731)

Example 2 Use the Poisson finite-difference method with $n = 6$, $m = 5$, and a tolerance of 10^{-10} to approximate the solution to

$$\frac{\partial^2 u}{\partial x^2}(x,y) + \frac{\partial^2 u}{\partial y^2}(x,y) = xe^y, \quad 0 < x < 2, \quad 0 < y < 1,$$

with the boundary conditions

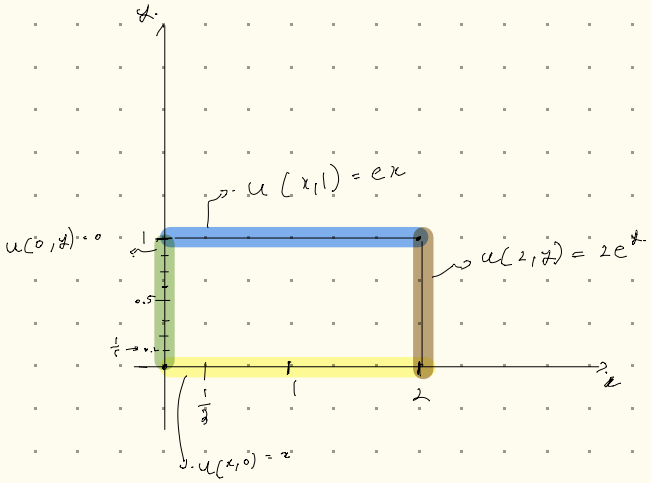
$$u(0,y) = 0, \quad u(2,y) = 2e^y, \quad 0 \le y \le 1,$$
$$u(x,0) = x, \quad u(x,1) = ex, \quad 0 \le x \le 2,$$

and compare the results with the exact solution $u(x,y) = xe^y$.

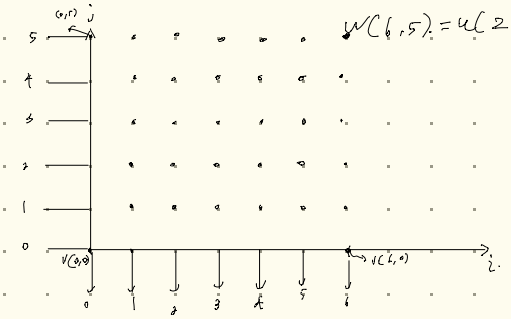
① $n=6$, $m=5$ grid 설정.

x, $0 \le x \le 2$ | y, $0 \le y \le 1$

$h = \frac{2}{n} = \frac{1}{3}$ | $k = \frac{1}{m} = \frac{1}{5}$



$W(i,j) = u(i \cdot h, j \cdot k)$



③ 이산화 과정.

$w(i,j) \approx u(i \cdot h, j \cdot k)$

$i : 0, 1, 2, 3, 4, 5, 6$

$j : 0, 1, 2, 3, 4, 5$

$h = \frac{1}{3}$ $k = \frac{1}{5}$

i 가 변화. $u(x,0)$, $u(0,y)$, $u(x,1)$, $u(2,y)$ 는 $w(i,j)$ 가 됨.

$W(0,0) = 0$

$u(2,0) = x$, $x = i \cdot h$

$W(1,0) = \frac{1}{3}$
$W(2,0) = \frac{2}{3}$
$W(3,0) = \frac{3}{3}$
$W(4,0) = \frac{4}{3}$
$W(5,0) = \frac{5}{3}$
$W(6,0) = \frac{6}{3}$

$u(0,y) = 0$, $y = j \cdot k$

$W(0,1) = 0$
$W(0,2) = 0$
$W(0,3) = 0$
$W(0,4) = 0$
$W(0,5) = 0$

$u(x,1) = ex$, $j \cdot \frac{1}{5} = 1$, $j = 5$, $x = i \cdot h$

$W(1,5) = \frac{e}{3}$
$W(2,5) = \frac{2e}{3}$
$W(3,5) = \frac{3e}{3}$
$W(4,5) = \frac{4e}{3}$
$W(5,5) = \frac{5e}{3}$
$W(6,5) = \frac{6e}{3}$

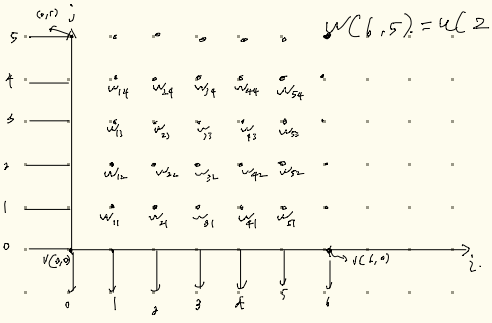
$u(2,y) = 2e^y$, $i \cdot \frac{1}{3} = 2$, $i = 6$, $y = j \cdot k$

$W(6,0) = 2$
$W(6,1) = 2e^{\frac{1}{5}}$
$W(6,2) = 2e^{\frac{2}{5}}$
$W(6,3) = 2e^{\frac{3}{5}}$
$W(6,4) = 2e^{\frac{4}{5}}$
$W(6,5) = 2e$

$1 \le i \le 5$, $1 \le j \le 4$

$w(i,j) \approx w_{ij}$

$w_{ij}^{(0)} = 0$ 초기값 0



② -2 차분법을 통한 푸아송 방정식의 전개.

▪ **Elliptic equation**

The *elliptic* partial differential equation we consider is the Poisson equation,

$$\nabla^2 u(x,y) \equiv \frac{\partial^2 u}{\partial x^2}(x,y) + \frac{\partial^2 u}{\partial y^2}(x,y) = f(x,y) \qquad (12.1)$$

$$\frac{\partial^2 u}{\partial x^2}(x_i,y_j) = \frac{u(x_{i+1},y_j) - 2u(x_i,y_j) + u(x_{i-1},y_j)}{h^2} - \frac{h^2}{12}\frac{\partial^4 u}{\partial x^4}(\xi_i,y_j), \text{ where } \xi_i \in (x_{i-1},x_{i+1}).$$

$$\frac{\partial^2 u}{\partial y^2}(x_i,y_j) = \frac{u(x_i,y_{j+1}) - 2u(x_i,y_j) + u(x_i,y_{j-1})}{k^2} - \frac{k^2}{12}\frac{\partial^4 u}{\partial y^4}(x_i,\eta_j), \text{ where } \eta_j \in (y_{j-1},y_{j+1}).$$

2계 편미분에 대한 차분법.

$\frac{\partial^2 u}{\partial x^2} \approx \frac{w_{i+1,j} - 2w_{ij} + w_{i-1,j}}{h^2}$ $\quad \frac{\partial^2 u}{\partial y^2} \approx \frac{w_{i,j+1} - 2w_{ij} + w_{i,j-1}}{k^2}$

푸아송 방정식에 적용.

$$\nabla^2 u(x,y) = \frac{\partial^2 u}{\partial x^2}(x,y) + \frac{\partial^2 u}{\partial y^2}(x,y) = f(x,y)$$

$$\approx \frac{w_{i+1,j} - 2w_{ij} + w_{i-1,j}}{h^2} + \frac{w_{i,j+1} - 2w_{ij} + w_{i,j-1}}{k^2} \Rightarrow -h^2 \cdot f(x,y) = 2w_{ij} - w_{i+1,j} - w_{i-1,j} + \frac{h^2}{k^2}w_{i,j+1} - \frac{h^2}{k^2}w_{i,j-1} - \frac{h^2}{k^2}w_{i,j+1}$$

$$= 2\left(1 + \left(\frac{h}{k}\right)^2\right)w_{ij} - \left(w_{i-1,j} + w_{i+1,j}\right) - \left(\frac{h}{k}\right)^2\left(w_{i,j-1} + w_{i,j+1}\right)$$

다음과 같이 $\nabla^2 u(x,y) = f(x,y)$ 에 대해서.

$2\left(1 + \left(\frac{h}{k}\right)^2\right)w_{ij} - \left(w_{i-1,j} + w_{i+1,j}\right) - \left(\frac{h}{k}\right)^2\left(w_{i,j-1} + w_{i,j+1}\right) = -h^2 \cdot f(x,y)$ 다음과 같이 나타낼 수 있다.

② -3 연립방정식 전개.

▪ **Finite Difference Method**

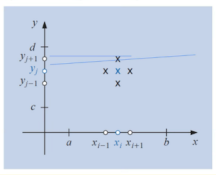
$$2\left[\left(\frac{h}{k}\right)^2 + 1\right]w_{ij} - (w_{i+1,j} + w_{i-1,j}) - \left(\frac{h}{k}\right)^2(w_{i,j+1} + w_{i,j-1}) = -h^2 f(x_i,y_j), \quad (12.4)$$

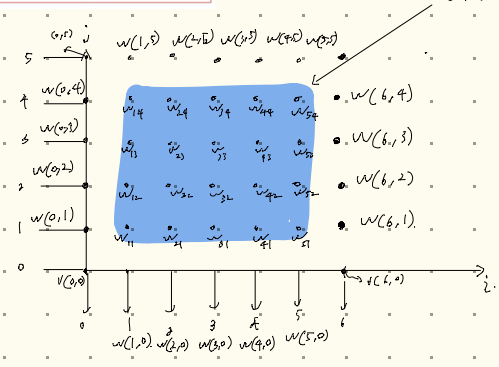
for each $i = 1,2,\ldots,n-1$ and $j = 1,2,\ldots,m-1$

The typical equation in (12.4) involves approximations to $u(x,y)$ at the points

(x_{i-1},y_j), (x_i,y_j), (x_{i+1},y_j), (x_i,y_{j-1}), and (x_i,y_{j+1}).

Figure 12.5





연립 방정식을 풀어 근사값을 구하려 하는데 변수 개수.

$h = \frac{1}{3}$ $\quad k = \frac{1}{5}$

$h^2 = \frac{1}{9}$ $\quad \frac{h^2}{k^2} = \frac{25}{9}$ $\quad 2\left(1 + \frac{h^2}{k^2}\right) = \frac{68}{9}$

$f(x_i,y_j) = x_i \cdot e^{y_j}$

$x_i = i \cdot h\left(=\frac{1}{3}\right)$ $\quad y_j = j \cdot k\left(=\frac{1}{5}\right)$

(i,j)

(1,1) $\Rightarrow \frac{68}{9}w_{11} - \left(w_{21} + w(0,1)\right) - \frac{25}{9}\left(w_{12} + w(1,0)\right) = \left(\frac{1}{3}\right)e^{\left(\frac{1}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(2,1) $\Rightarrow \frac{68}{9}w_{21} - \left(w_{31} + w_{11}\right) - \frac{25}{9}\left(w_{22} + w(2,0)\right) = \left(\frac{2}{3}\right)e^{\left(\frac{1}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(3,1) $\Rightarrow \frac{68}{9}w_{31} - \left(w_{41} + w_{21}\right) - \frac{25}{9}\left(w_{32} + w(3,0)\right) = \left(\frac{3}{3}\right)e^{\left(\frac{1}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(4,1) $\Rightarrow \frac{68}{9}w_{41} - \left(w_{51} + w_{31}\right) - \frac{25}{9}\left(w_{42} + w(4,0)\right) = \left(\frac{4}{3}\right)e^{\left(\frac{1}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(5,1) $\Rightarrow \frac{68}{9}w_{51} - \left(w(6,1) + w_{41}\right) - \frac{25}{9}\left(w_{52} + w(5,0)\right) = \left(\frac{5}{3}\right)e^{\left(\frac{1}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(1,2) $\Rightarrow \frac{68}{9}w_{12} - \left(w_{22} + w(0,2)\right) - \frac{25}{9}\left(w_{13} + w_{11}\right) = \left(\frac{1}{3}\right)e^{\left(\frac{2}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(2,2) $\Rightarrow \frac{68}{9}w_{22} - \left(w_{32} + w_{12}\right) - \frac{25}{9}\left(w_{23} + w_{21}\right) = \left(\frac{2}{3}\right)e^{\left(\frac{2}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(3,2) $\Rightarrow \frac{68}{9}w_{32} - \left(w_{42} + w_{22}\right) - \frac{25}{9}\left(w_{33} + w_{31}\right) = \left(\frac{3}{3}\right)e^{\left(\frac{2}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(4,2) $\Rightarrow \frac{68}{9}w_{42} - \left(w_{52} + w_{32}\right) - \frac{25}{9}\left(w_{43} + w_{41}\right) = \left(\frac{4}{3}\right)e^{\left(\frac{2}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(5,2) $\Rightarrow \frac{68}{9}w_{52} - \left(w(6,2) + w_{42}\right) - \frac{25}{9}\left(w_{53} + w_{51}\right) = \left(\frac{5}{3}\right)e^{\left(\frac{2}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(1,3) $\Rightarrow \frac{68}{9}w_{13} - \left(w_{23} + w(0,3)\right) - \frac{25}{9}\left(w_{14} + w_{12}\right) = \left(\frac{1}{3}\right)e^{\left(\frac{3}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(2,3) $\Rightarrow \frac{68}{9}w_{23} - \left(w_{33} + w_{13}\right) - \frac{25}{9}\left(w_{24} + w_{22}\right) = \left(\frac{2}{3}\right)e^{\left(\frac{3}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(3,3) $\Rightarrow \frac{68}{9}w_{33} - \left(w_{43} + w_{23}\right) - \frac{25}{9}\left(w_{34} + w_{32}\right) = \left(\frac{3}{3}\right)e^{\left(\frac{3}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(4,3) $\Rightarrow \frac{68}{9}w_{43} - \left(w_{53} + w_{33}\right) - \frac{25}{9}\left(w_{44} + w_{42}\right) = \left(\frac{4}{3}\right)e^{\left(\frac{3}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(5,3) $\Rightarrow \frac{68}{9}w_{53} - \left(w(6,3) + w_{43}\right) - \frac{25}{9}\left(w_{54} + w_{52}\right) = \left(\frac{5}{3}\right)e^{\left(\frac{3}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(1,4) $\Rightarrow \frac{68}{9}w_{14} - \left(w_{24} + w(0,4)\right) - \frac{25}{9}\left(w(1,5) + w_{13}\right) = \left(\frac{1}{3}\right)e^{\left(\frac{4}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(2,4) $\Rightarrow \frac{68}{9}w_{24} - \left(w_{34} + w_{14}\right) - \frac{25}{9}\left(w(2,5) + w_{23}\right) = \left(\frac{2}{3}\right)e^{\left(\frac{4}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(3,4) $\Rightarrow \frac{68}{9}w_{34} - \left(w_{44} + w_{24}\right) - \frac{25}{9}\left(w(3,5) + w_{33}\right) = \left(\frac{3}{3}\right)e^{\left(\frac{4}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(4,4) $\Rightarrow \frac{68}{9}w_{44} - \left(w_{54} + w_{34}\right) - \frac{25}{9}\left(w(4,5) + w_{43}\right) = \left(\frac{4}{3}\right)e^{\left(\frac{4}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

(5,4) $\Rightarrow \frac{68}{9}w_{54} - \left(w(6,4) + w_{44}\right) - \frac{25}{9}\left(w(5,5) + w_{53}\right) = \left(\frac{5}{3}\right)e^{\left(\frac{4}{5}\right)} \cdot \left(-\frac{1}{3}\right)^2$

20개 미지수에 대한 20개의 종속 관계 사이의 연립방정식을 가우스 소거법을 통해 해를 구할 수 있음.

② 가우스-사이델 이용한 근사 해도출.

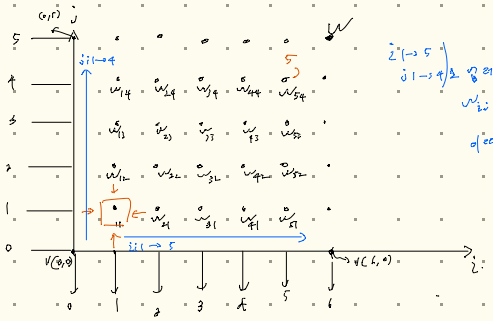
i: 1, 2, 3, 4, 5 j: 1, 2, 3, 4 에서의

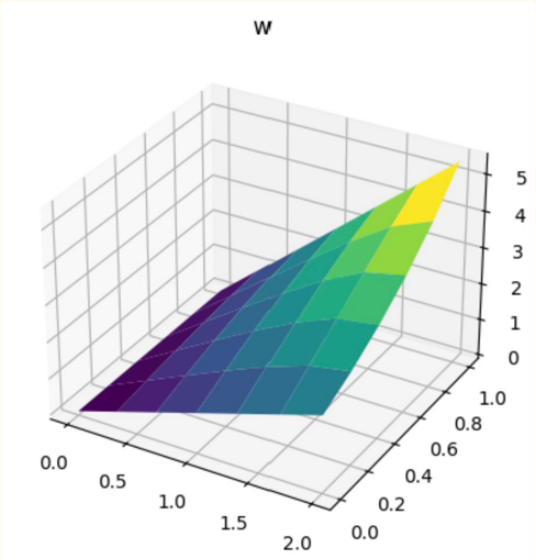
$\frac{4h}{q} w_{ij} - \left(w_{i+1,j} + w_{i-1,j}\right) - \frac{25}{q}\left(w_{i,j+1} + w_{i,j-1}\right) = \left(\frac{1}{5}i\right) e^{\left(\frac{1}{5}ij\right)}$

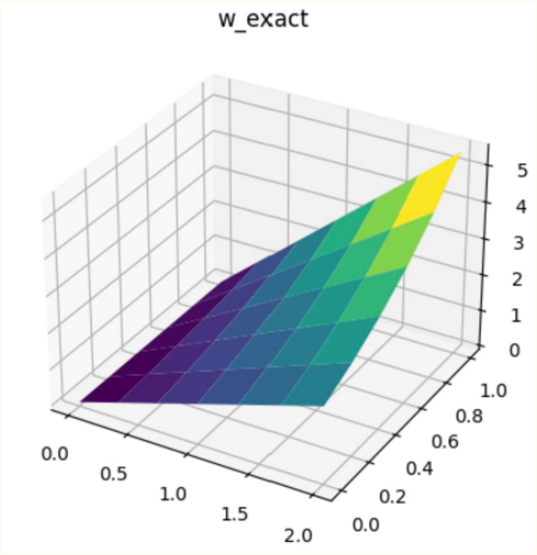
에 대하여.

$$w_{ij} = \frac{1}{2\left(1+\left(\frac{h}{u}\right)^2\right)}\left(-h^2 \cdot f(x,y) + \left(w_{i-1,j} + w_{i+1,j}\right) + \left(\frac{h}{u}\right)^2\left(w_{i,j+1} + w_{i,j-1}\right)\right).$$

$$w_{ij} = \frac{q}{4h}\left(-\left(\frac{1}{5}\right)^3 (ij)\, e^{\left(\frac{1}{5}ij\right)} + \left(w_{i+1,j} + w_{i-1,j}\right) + \frac{25}{q}\left(w_{i,j+1} + w_{i,j-1}\right)\right)$$



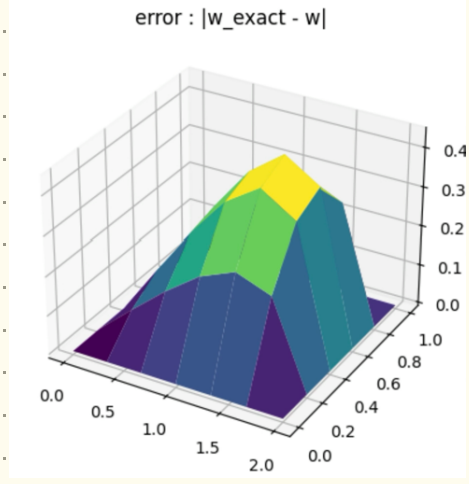





```
w
[[0.         0.         0.         0.         0.         0.        ]
 [0.33333333 0.48537807 0.62232321 0.74130446 0.83793189 0.90609394]
 [0.66666667 0.9674649  1.23929585 1.47721854 1.67250726 1.81218789]
 [1.         1.43954552 1.83995567 2.19659043 2.49669504 2.71828183]
 [1.33333333 1.88462712 2.39645826 2.87053445 3.29170583 3.62437577]
 [1.66666667 2.25838066 2.83682645 3.4214938  4.00284429 4.53046971]
 [2.         2.44280552 2.9836494  3.6442376  4.45108186 5.43656366]]
w_exact
[[0.         0.         0.         0.         0.         0.        ]
 [0.33333333 0.40713425 0.4972749  0.60737293 0.74184698 0.90609394]
 [0.66666667 0.81426851 0.9945498  1.21474587 1.48369395 1.81218789]
 [1.         1.22140276 1.4918247  1.8221188  2.22554093 2.71828183]
 [1.33333333 1.62853701 1.9890996  2.42949173 2.9673879  3.62437577]
 [1.66666667 2.03567126 2.4863745  3.03686467 3.70923488 4.53046971]
 [2.         2.44280552 2.9836494  3.6442376  4.45108186 5.43656366]]
```

w: 근사해 (가우스-사이델; N: 100 Tol: 1e-6).

w: xe^y, $x = i \cdot h$, $y = j \cdot k$ 에 의해 계산된 값.




```
error
[[0.         0.         0.         0.         0.         0.        ]
 [0.         0.07824381 0.12504831 0.13993152 0.09608492 0.        ]
 [0.         0.1531964  0.24474605 0.26247267 0.1888133  0.        ]
 [0.         0.21814276 0.34813097 0.37447163 0.27115411 0.        ]
 [0.         0.25609011 0.40735866 0.44104272 0.32431792 0.        ]
 [0.         0.2227094  0.35045195 0.38462913 0.29360941 0.        ]
 [0.         0.         0.         0.         0.         0.        ]]
```

다음과 같이 정확한 값에 접근한다.

유한차분법의 오차 $O(h^2, k^2)$ 로 인한 수렴결과를

오차가 커짐을 확인할 수 있다.

작성코드

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
a = 0.0
b = 2.0
c = 0.0
d = 1.0

n = 6
m = 5
h = (b-a) / n
k = (d-c) / m
TOL =  1e-6
N = 100

x = np.linspace(a , b, n + 1)
y = np.linspace(c , d, m + 1)

xx, yy = np.meshgrid(x, y, indexing='ij')
w = np.zeros((n + 1, m + 1))


#초기값 설정
w[:,0] = x
w[0,:] = 0.0

w[:,-1] = np.exp(1) * x
w[-1,:] = 2 * np.exp(y)

w_exact = w.copy()

it = 0
err = 1.0
while it < N and err > TOL:
  max_err = 0
  for i in range(1 , n):
    for j in range(1 , m):
      w_new = (1/( 2*(1+(h/k)**2) ))* ( (-h)**2 * (i*h) * np.exp((k*j)) + w[i+1 ,j] + w[i-1 ,j] + (h/k)**2 * (w[i ,j+1] + w[i ,j-1]) )
      max_err = max(max_err , abs(w_new - w[i,j]))
      w[i , j] = w_new
      w_exact[ i , j] = (i*h) * np.exp(j*k)

  err = max_err
  it += 1
print("w")
print(w)
print("w_exact")
print(w_exact)
print("error")
print(abs(w - w_exact))

fig = plt.figure()
ax = fig.add_subplot(projection='3d')
ax.set_title("w")
surf = ax.plot_surface(xx , yy, w , cmap= 'viridis')

fig = plt.figure()
ax = fig.add_subplot(projection='3d')
ax.set_title("w_exact")
surf = ax.plot_surface(xx , yy, w_exact , cmap= 'viridis')

fig = plt.figure()
ax = fig.add_subplot(projection='3d')
ax.set_title("error : |w_exact - w|")
surf = ax.plot_surface(xx , yy, abs(w - w_exact) , cmap= 'viridis')
```